Data Warehouse Design & Architecture

1. Architecture Approach

We selected a **Bottom-Up (Kimball) approach** using a **Star Schema**. Data is extracted from three heterogeneous sources (MySQL, Excel, CSV), loaded into a Python/Pandas staging area for cleansing, and finally loaded into the centralized MySQL Data Warehouse.

2. Dimensional Model (Star Schema)

- **Fact Table:** `library_usage_fact` (Contains measures: `download_count`, `duration_minutes`, `loan_days`, `duration_hours`).
- **Dimension Tables:**
  - `dim_date`: Tracks year, quarter, month, day.
  - `dim_student`: Tracks department and user type.
  - `dim_resource`: Tracks resource type (Physical Book, E-book, Journal) and category.
  - `dim_room`: Tracks room numbers.
  - `dim_time`: Tracks time slots (Morning, Afternoon, Evening).

3. Justification for Star Schema

We chose a Star Schema over a Snowflake Schema because it is highly denormalized, which significantly improves query performance for OLAP operations (like slicing and dicing). It is also much easier for the Library Director to understand and use for generating reports.

4. Handling Date Inconsistencies

Because the three source systems used different date formats (e.g., "Jan 15, 2024", "2024-01-15"), we created a `dim_date` table. During the ETL phase, all dates were parsed using Python's `pd.to_datetime` and standardized to ISO 8601 (YYYY-MM-DD) before generating a `date_key` (e.g., 20240115) to link to the fact table.